



**UNIVERSIDAD AUTÓNOMA
DE AGUASCALIENTES**

Project proposal – Gridiron Heatmaps

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GitHub repository link: <https://github.com/JoaoSanBills/Gridiron-Heatmaps>

Technology stack

- **Frontend:** HTML5, CSS3 (Flexbox/Grid for interfaces) y JavaScript (Vanilla, ES6+ for DOM and graphics)
- **Backend:** Node.js (Express.js) or using an existing API
- **Database:** JSON files structured and managed by the Node.js server, we can have the backend using an external NFL data API (via Axios or Fetch) to process everything before sending it to the frontend
- **Libraries & dependencies:** Express and Cors configured for the backend server, HTML5 Canvas API on the frontend to render the interactive field and heat map

Description

A web analytics platform with a Client-Host architecture that visualize the offensive and defensive tendencies in the NFL through heat maps (Hot Zones), the backend, developed on Node.js it will handle the complex logic: processing, filtering and structuring the statistic records of the quarterbacks (QBs) and wide receivers (WRs), the frontend consumes the server's API and presents an interactive graphic field where, upon selecting a player under specific game scenarios (e.g. 3rd down and 5 yards to go), it dynamically renders the data by painting the field zones to indicate efficiency percentages and preferred routes.

What problem does the project aim to solve and why is important?

The Problem: Processing massive databases of sports statistics directly in the user's browsers (frontend) make applications slow and causes them to freeze, besides, for analysts or fans, interpreting giant numerical tables to find trends is confusing and inefficient

The Importance: By moving data processing to the backend with Node.js, we radically optimize the application's performance and separate responsibilities, it instantly translates abstract data into visual intelligence, professionally, it demonstrates our ability to build a complete Full stack solution: this will help us know how to configure a server, create endpoints (APIs), process complex structures on the backend, and consume that data asynchronously to render an advanced graphical interface